Matthieu Huguet

Technical Artist - Freelance

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Aix-en-Provence, France · Remote Open to contract & freelance remote · min. 3 months

Technical Artist specializing in Houdini + Unreal Engine 5. From procedural modeling to pipeline architecture - I design the systems studios need to scale production, automate workflows, and ship environments at a high visual standard. Tool development in Python, C++ and MCP. Bachelor Game Design LISAA Paris (2023) · Gnomon Online · CS50 Harvard (2024).

Technical Skills

ENGINES / TOOLS

Unreal Engine 5 ●●●●●
`Blueprint` `PCG` `Material/Shader` `UMG` `Niagara`

3D / PROCEDURAL

Houdini ●●●●●
`VEX` `HDAs` `Procedural` `UV` `FLIP` `Heightfields`

Maya ●●●●●
`Rigging` `Hard Surface` `Animation` `Modeling`

Substance Designer ●●●●○
`Procedural PBR` `Node Graph` `MCP Plugin`

Substance Painter ●●●●○
`Texturing` `Baking` `PBR`

ZBrush ●●●○○
`Sculpting` `Organic`

SpeedTree ●●●○○
`Vegetation` `Foliage`

PIPELINE / CODE

Claude Code / Cursor ●●●●○
`AI Coding` `Agent Workflows` `Automation Pipelines` `MCP` `.md Memory Files` `Instruction Prompts`

Python ●●●●○
`Automation Pipelines` `Tools` `PushVault` `MCP`

RizomUV ●●●●○
`UV Unwrap` `UDIM`

Adobe Suite ●●●●○
`Photoshop` `Premiere` `InDesign`

C++ / UE5 ●●●○○
`Plugin Dev` `Editor Extension`

React / JavaScript ●●●○○
`Web Tools` `Portfolio`

Figma / Miro ●●●○○
`UI/UX` `Prototype` `Wireframe`

Documentation ●●●○○
`Word` `Google Docs`

Languages

French ●●●●● Native — English ●●●●○ Fluent (C1+) — Japanese ●●○○○ Intermediate (N4)

Projects - matthieuhuguet.com

ArtStation ↗

Fleur de Lys: Procedural Lily, Houdini to Unreal ↗

Houdini · Copernicus · UE5

Procedural lily built as a Houdini HDA with a Copernicus trim sheet and rendered in Unreal Engine 5.8 with Lumen.

`Houdini` `HDA` `Copernicus` `UE5` `Lumen`

Ceramic Tiles - Made with Houdini Copernicus and Unreal Lumen ↗

Houdini · Copernicus · UE5

Ceramic tile material rebuilt in Houdini Copernicus after Julien Delannoy's Substance 3D Designer project, rendered in Unreal Engine 5.8 with Lumen.

`Copernicus` `Texturing` `UE5` `Lumen`

HDA Stairs Component

Houdini HDA

Procedural staircase HDA for the Louisiana Bayou Church. 37 nodes, 3 exposed parameters: height, step count, tread depth. Stringers, treads, pillars, rails and rungs, mirrored for bilateral symmetry.

`Houdini` `HDA` `Procedural` `Architecture`

UV Texture Variation Offset

Houdini

Procedural UV via copytopoints + name matching. 64MB → 16MB texture memory.

`VEX` `UV` `Pipeline` `Optimization`

Louisiana Bayou Church

Houdini

Red Dead-style wooden church. Modular 6-component HDA (Armature, WoodFacade, Roof, Window x2, Pillars) with exposed parameters.

`Procedural` `Architecture` `HDAs`

Procedural Shutters in Houdini

Houdini

Procedural shutter asset for a larger procedural house system, built on top of existing geometry groups to avoid breaking the model.

`Houdini` `Procedural` `HDA`

Hero Asset Texturing Pipeline

Houdini · RizomUV · Substance · UE5

Full pipeline: geometry repair (Houdini), automated UDIMs (RizomUV), Megascans database (Substance Designer), Virtual Texturing (UE5).

`Pipeline` `UDIM` `Substance` `UE5`

Asset Scattering & Placement

Houdini · UE5

UE5 PCG replacement via single HDA. Heightfield terrain erosion, 3-pass vegetation, Labs Pick and Place, 8-bit Dither TAA water mask.

`Houdini` `Level Design` `PCG` `Heightfield`

Procedural Grass System

Houdini · Substance · UE5

Full procedural grass pipeline. Dual-noise VAT wind animation, seamless crossfade - inspired by Ghost of Tsushima.

`Houdini` `Substance` `VAT` `Vegetation`

Material Layer System

Unreal Engine 5

Modular material pipeline - Material Layers + Vertex Color blending. 188 shader instructions. Nanite + ray tracing compatible. RGBA mask bake option.

`Shader` `Material` `UE5` `Pipeline`

Louisiana Bayou V1.5

Unreal Engine 5

Atmospheric environment - sculpted SpeedTree cypress, Single Layer Water, Blueprint Spline system, multi-layer PBR texturing.

`UE5` `Environment` `SpeedTree` `Shader`

Texturing Cliff

Substance Designer

Procedural sedimentary rock material. Erosion, cracks, color variation. Custom sub-graphs, chunk breakup, fuzzy edges. 4K.

`Substance` `Procedural` `Texturing`

Substance Ornate Fabric Introduction

Substance Designer

First fabric material study: ornate woven pattern with procedural thread structure, weave variation and surface relief. Built from scratch: fiber direction, interlace logic, per-thread color variation, micro-detail normal pass.

`Substance` `Procedural` `Texturing` `Fabric`

Houdini Game Environments

Houdini · Maya · ZBrush · UE5

Study of Creating Assets and Architecture for Game Environments (Giovani Magana), from Maya and ZBrush to Unreal Engine 5. Procedural modeling HDA driven by the normal attribute for orientation and alignment, with quick variation of dimensions, materials and ornamentation.

`Houdini` `HDA` `Modularity` `Environment`

Water Wheel

Houdini

Procedural water wheel geometry + FLIP water simulation. Parametric: size, bucket count, wood aging, moss growth.

`Houdini` `FLIP` `Procedural` `Simulation`

Niagara Vortex VFX

Niagara · Unreal Engine 5

Niagara and custom meshes for magical vortex effects that respond to gameplay in real time. Interactive fur simulation, vortex forces, stylized rendering.

`Niagara` `VFX` `UE5` `Grooming`

Procedural Rock Gen

Houdini · Point VOP

Rock sculpting workflow using Point VOP with controlled noise displacement. Pipeline: cube to VDB from poly, smooth, convert. Realistic weathering and surface detailing.

`Houdini` `VOP` `Procedural` `Sculpting`

Pampa Grass

Houdini HDA · UE5

Modular vegetation HDA. Architecture: fiber, column, stems, grass. Vertex color SSS + wind mask. Infinite variations, LOD support.

`Houdini` `VEX` `HDA` `Vegetation` `UE5`

ZenRayProgress Design Tour

UX / UI · React

Component-by-component tour of the ZenRayProgress design, screen by screen: drive, calendar, timer, notebook and library in one web app.

`UX/UI` `Design System` `React`

MetaHuman Markerless macOS Plugin

Unreal Engine · MetaHuman

Markerless motion capture on macOS with the MetaHuman Animator plugin: a four second attack animation from a gameplay video, retargeted onto SKM_Quinn in Unreal Engine 5. Runs locally, 1080p to 2160p.

`UE5` `MetaHuman` `Mocap` `macOS`

ZenRay Dictate App

macOS App

macOS dictation that works everywhere, built from ChatGPT's dictation component: Fn brings up a small bar, Cmd+D starts and stops, the text lands on the clipboard. Free, open source (MIT).

`macOS` `Dictation` `Open Source`

ZenRayProgress App

App · React

Personal drive in a tab: one app that centralizes my files and lets me open and edit them. Preview build, still being enhanced.

`React` `Web App` `Tools`

MagickSlides

Python · ImageMagick

Premium slide generation via ImageMagick + code execution. Built with Cursor IDE + Gemini Agent.

`Python` `Tool` `Automation`

Substance Designer MCP Plugin

GitHub ↗

MCP · Python

MCP plugin for Substance Designer. Natural language → complete functional PBR graph. Open source.

`MCP` `Python` `AI` `Substance` `Open Source`

PushVault

GitHub ↗

Python Tool

Automatic multi-repo backup utility to GitHub. Daily branches, batch push 50 files, diff viewer. Open source.

`Python` `Pipeline` `Open Source` `GitHub`

UE5 MCP Plugin

C++ · Unreal Engine 5

C++ UE5.7 plugin - AI agents manipulate the editor via MCP (scene query, spawn actors, Blueprint generation). Claude Code + Cursor compatible.

`C++` `UE5` `MCP` `AI` `Plugin`

Final Fantasy VII Game UI Recreation

React · UE5

React tool to recreate FF7R UI. Design > animation > export. Clean component architecture.

`React` `UI/UX` `Game UI` `Tool`

Ghost of Tsushima Game UI Recreation

Unreal Engine 5

Sucker Punch micro-motion study. Curve animations, Material Parameter Collections sync UI/Post-Process, clean UMG/Blueprint architecture.

`UE5` `UMG` `Blueprint` `UI/UX`

Noyes & LISAA Games 2021-2023

Unity · LISAA Paris

Trailer compilation of 7 games. Lead GD & Programmer on Noyes (graduation project, atmospheric narrative game), Sea Of Love, Beat'em Up, Juryoku To, Turtle Quest, Crazy Kamikaze, Koldstare.

`Unity` `Game Design` `Narrative` `Lead`

Experience

Oct 2025 - Nov 2025
Freelance

Technical Artist & Designer — Meerkat Studio — Remote

- Unannounced UE5 project (NDA) - production-ready prototype delivered.
- Game Design Pillars architecture and full GDD structured in Notion.
- Houdini-to-UE5 workflow for fast, scalable environment set dressing.
- Modular Blueprint mechanics with Data Assets for flexible architecture.
- Reactive Niagara VFX for core loop validation.
- Complete simulation game GDD (Animal Shelter style) - game loop, progression, documentation.

Feb 2025 - Nov 2025
Freelance

Technical Artist — ZenRay — Remote

- Environments, prototypes, GDDs, Archviz, Game Ready Assets, Architecture.
- Archviz client project - high-fidelity 3D reproduction of a property for sale, Aix-en-Provence.
- Procedural systems via Houdini HDAs - vegetation, terrain, scatter.
- PBR materials and texturing via Substance 3D Designer and Painter.
- Open source MCP plugins published: UE5 editor + Substance Designer.

2025
Freelance

Archviz 3D — Client Particulier — Aix-en-Provence

- High-fidelity 3D reproduction of a real estate property for sale.
- Deliverables: renders, turntable, virtual walkthrough.

Education

Feb - Nov 2024

Procedural & Technical Art Specialisation — Gnomon Online

Procedural Workflows for Games (Houdini), Creating Environment Art (Ciro Cardoso), Introduction to ZBrush (Madeleine Scott-Spencer), Introduction to Maya Vol.1-2-3 (Eric Keller), Creating a Monolithic City in UE5 (Jianfeng Li), UE5 Material System Overview, Creating Assets & Architecture for Game Environments (Giovani Magana), Intro Substance Painter (Christophe Desse), Real-Time Filmmaking in UE5, and more.

`Houdini` `ZBrush` `Maya` `UE5` `Substance` `Environment Art` `Rigging`

Feb - Jul 2024

CS50: Introduction to Computer Science — Harvard Online

Scratch, C, Python, SQL, HTML/CSS/JavaScript, Flask, AI, data structures, memory, algorithms, cybersecurity. Prof. David J. Malan. Followed by Introduction to Game Development.

`C` `Python` `SQL` `JavaScript` `Algorithmes`

Sep 2023 - Feb 2024

Master Game Design & Development, Game Art — ARTFX Montpellier

Courses attended, then self-directed path for more targeted technical art skills.

`Houdini` `Rigging` `UE5` `Environment Art`

2020 - 2023

Bachelor Game Design — LISAA Paris

2022-2023 - Lead GD & Programmer: Noyes (graduation project, Unity), Gravity Tower (advanced physics), Beat'em Up Project (combat design). 2021-2022 - GD & Programmer: Game Jams Turtle Quest & Captain Sveltana, Pawfee Shop (AI & programming), Versus Games Crazy Kamikazee & Koldstare. Multidisciplinary team leadership.

`Unity` `Game Design` `Level Design` `UI/UX` `Adobe Suite`

Other

Aug - Sep 2023

Foundation in Mediation & Negotiation — IHEMN

Institut des Hautes Études en Médiation et Négociation - conflict resolution processes, advanced negotiation.

Jun - Sep 2023

Introduction to Rigging & Maya for Beginners — Udemy

Complete Guide to 3D Animation in Maya. Instructors: Abraham Leal, Phil Ebiner, Lucas Ridley.

2019 - 2020

Preparatory Class for Grandes Écoles (Humanities) — Lycée Faidherbe, Lille

Philosophy, History, Geography, English, Italian, Classical Languages.

2018 - 2019

Regional Youth Delegate — Région Sud

Youth civic mandate under President Renaud Muselier - regional representation, terminal year.

2016 - 2019

Baccalauréat, Literature track - Honours — Lycée Lacordaire, Marseille

Options: Mathematics & Visual Arts.